

it can get us into problems 10-20 years down line,” says Mangal.

For Tata Steel, the sustainability push is not centred around a single technology bet. Instead, the company is pursuing multiple pathways. “Our approach is a combination of deploying available technologies and investing for the future—from increasing renewable energy use and energy-efficiency projects to exploring biochar as a substitute for coal and increasing scrap usage in steelmaking,” he says.

That broader strategy spans energy-efficient technologies, embedding climate risk into enterprise decision-making, biodiversity conservation and circular economy initiatives. The company has aligned itself with the World Economic Forum’s

46 | Trillion Trees initiative through afforestation targets, while also pushing towards complete utilisation of industrial by-products.

Circularity, in fact, has evolved into a business vertical of its own within Tata Steel. Mangal says the company prefers the term “by-product management” over waste management because industrial waste can become a revenue-generating asset. At its current production scale of nearly 25 million tonnes, Tata Steel generates around 15 million tonnes of by-products annually.

“Many years ago, we realised that what is often treated as waste can be converted into valuable products,” he says. The company subsequently created a dedicated Industrial By-Product Management Division focused on extracting commercial value from industrial waste.

At present, 99.9% of Tata Steel’s by-products are either recycled, re-

used or consumed by other industries, with less than 0.1% going to the landfill, says Mangal. The company is aiming for full utilisation of by-products by 2030.

Water conservation has also emerged as a central pillar of Tata Steel’s sustainability road map, particularly as many of its operations are in water-stressed regions. The company has set a target to become water positive by 2030.

Tata Steel currently operates across nine river basins in India owing to its presence across eastern, northern, western and southern geographies. According to Mangal, the company has already achieved water positivity in five river basins, while work is underway in the remaining four.

Even as Tata Steel ramps up investments in low-carbon technologies, Mangal acknowledges that the economics of green steel remain challenging due to higher costs.

The company is engaging with policymakers on several fronts and advocating government-led green procurement, where infrastructure projects such as railways and bridges prioritise low-carbon steel and create a premium market for sustainable products. Beyond that, Tata Steel sees long-term opportunities in clean energy ecosystems such as hydrogen and nuclear power, areas where policy clarity and public-private collaboration could shape the future of industrial decarbonisation in India.

For Tata Steel, it is increasingly becoming a long-term business strategy, one that seeks to prepare the company for a future where growth, climate accountability and industrial competitiveness coexist. **BT**

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MOST SUSTAINABLE COMPANY IN MANUFACTURING (METALS & MINING)

○
Large-scale deployment
of energy-efficient
technologies

○
Climate risk embedded
into enterprise
management framework

○
Strong commitment
towards “no net loss”
biodiversity approach

○
Expansion of circular
economy practices (slag
utilisation, resource
efficiency)

○
100% utilisation of
by-products (slag, fly
ash) through industrial
symbiosis

○
Advanced water
stewardship practices at
multiple sites

